Exercise 21.2

After Fred receives the phone-call about the burglar alarm, but before he hears the radio report, what, from his point of view, is the probability that there was a small earthquake today?

Solution:

$$P(e = 1|a = 1) = 0.006$$

Exercise 21.3

Imagine fitting a mixture of ten Gaussians to data in a twenty-dimensional space. Estimate the computational cost of implementing inferences for this model by enumeration of a grid of parameter values.

Solution:

The number of free parameters are 20*10=200 for mean, 20*10=200 for variance and 10-1=9 for the mixing weights, giving 409 in total. This mean if we grid search with x points per parameter, the computational cost will be $O(x^{409})$.